Yegor Lushpin

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WORK EXPERIENCE

BEARCATS ELECTRIC RACING **Aug. 2025 – Present**
Accumulator & Powertrain — FSAE EV *Cincinnati, OH*

- Supported the design and integration of the high-voltage accumulator (battery pack), drafting busbar and fusible-link drawings for overcurrent protection in compliance with FSAE rules and electrical standards
- Fabricated fusible links and validated their current-carrying dimensions for design research using an oscilloscope, characterizing fault behavior to confirm reliable overcurrent protection across the pack
- Assembled BMS sense boards that interface directly with the accumulator's cell stack, providing per-cell voltage and temperature monitoring critical to safe high-voltage operation
- Designed a cell temperature-monitoring system to validate the thermal performance of the battery pack under load, ensuring cells stayed within safe operating limits during testing
- Built a real-time CAN bus data acquisition system on a Raspberry Pi (MCP2515 CAN HAT) streaming decoded pack telemetry — voltages, temperatures, and state-of-charge — at 500 kbps to a cloud time-series database (InfluxDB) via Python and DBC decoding

SUNSKETCHER **Feb. 2024 – Aug. 2025**
Junior iOS Developer — NASA-Supported App *Bowling Green, KY*

- Contributed to a NASA-supported citizen-science app that captures total solar eclipse imagery, helping researchers measure the Sun's size, shape, and internal dynamics
- Developed encrypted server backup for the iOS application in Swift, securing user-contributed eclipse data in transit and at rest across the project's collection pipeline
- Modeled eclipse simulations in MATLAB from collected field data, collaborating with NASA engineers and leading heliophysicists to refine timing accuracy

PERSONAL HARDWARE BUILDS **2024 – Present**
Embedded Systems & Firmware *Cincinnati, OH*

- Designed and built portable embedded devices on ESP32-S3 (PlatformIO/C++), integrating I2C/I2S sensors and OLED displays with custom pin planning, paired with a companion Python stack (Flask, Whisper, local LLM)
- Designed schematics and PCBs in KiCad, including a long-range LoRa radio link for off-grid communication across the Tristate Area.

WENDY'S **Mar. 2025 – Present**
Crew Member *Alexandria, KY*

- Delivered fast, friendly service in a high-volume setting while maintaining a consistent 97–99% store cleanliness score; awarded the store's first-ever Employee of the Month

EDUCATION

University of Cincinnati **May 2030**
B.S., Electrical Engineering *Cincinnati, OH*

Campbell County High School **Jun. 2025**
Summa Cum Laude, 4.00 GPA *Alexandria, KY*

ACTIVITIES & LEADERSHIP

TECHNOLOGY STUDENT ASSOCIATION **Aug. 2023 – May 2025**
Team Leader *Alexandria, KY*

- Led a team to plan, build, and present Engineering, Coding, and CAD projects at regional, state, and national TSA competitions, coordinating deadlines and mentoring teammates

SKILLS & INTERESTS

- **Languages:** Python, C/C++, Java, MATLAB, Swift
- **Tools:** KiCad, LabVIEW, Git, Linux, PlatformIO, Visual Studio Code, Microsoft Office
- **Hardware:** Oscilloscope, multimeter, soldering, CAN bus, I2C/I2S, microcontrollers (ESP32, Raspberry Pi)
- **Interests:** hiking, creative writing, radio communications, reading, web development, weight lifting